

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12406784
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Maehara; Hisashi

Method of producing bonded magnet comprising SmFeN-based anisotropic magnetic powder

Abstract

A method of producing a SmFeN-based anisotropic magnetic powder is provided, the method including preparing a SmFeN-based anisotropic magnetic powder before dispersing comprising Sm, Fe, W, and N, and dispersing the SmFeN-based anisotropic magnetic powder before dispersing using a resin-coated metal media or a resin-coated ceramic media to obtain a SmFeN-based anisotropic magnetic powder. Also provided is a SmFeN-based anisotropic magnetic powder comprising Sm, Fe, W, and N and having an average particle size of less than 2.5 μm , a residual magnetization σ_r of not less than 130 emu/g, and an oxygen content of not higher than 0.75% by mass.

Inventors:	Maehara; Hisashi (Itano-gun, JP)
Applicant:	NICHIA CORPORATION (Anan, JP)
Family ID:	84363521
Assignee:	NICHIA CORPORATION (Anan, JP)
Appl. No.:	17/806376
Filed:	June 10, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220406496 A1	Dec. 22, 2022

Foreign Application Priority Data

JP	2021-097489	Jun. 10, 2021
JP	2022-088565	May. 31, 2022

Publication Classification

Int. Cl.: **H01F1/059** (20060101); **B22F9/04** (20060101); **C22C33/02** (20060101); **C22C38/00** (20060101)

U.S. Cl.:

CPC **H01F1/059** (20130101); **C22C33/0235** (20130101); **C22C38/001** (20130101); **C22C38/005** (20130101); **C22C2202/02** (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4692420	12/1986	Oda	501/153	C04B 35/584
6328817	12/2000	Murakami	N/A	N/A
2005/0076974	12/2004	Honkura	148/301	H01F 1/061
2006/0022175	12/2005	Komuro et al.	N/A	N/A
2006/0191601	12/2005	Komuro et al.	N/A	N/A
2008/0035879	12/2007	Matsukawa et al.	N/A	N/A
2010/0289366	12/2009	Komuro et al.	N/A	N/A
2011/0240909	12/2010	Kanda et al.	N/A	N/A
2012/0145944	12/2011	Komuro et al.	N/A	N/A
2012/0175986	12/2011	Komuro et al.	N/A	N/A
2013/0011660	12/2012	Diekmann et al.	N/A	N/A
2013/0273189	12/2012	Yamao	106/38.27	C01B 21/068
2013/0278104	12/2012	Komuro et al.	N/A	N/A
2017/0271928	12/2016	Sanada et al.	N/A	N/A
2018/0226180	12/2017	Kawashita et al.	N/A	N/A
2020/0016663	12/2019	Okada	N/A	H01F 1/059
2020/0098496	12/2019	Kinoshita et al.	N/A	N/A
2021/0098161	12/2020	Maehara	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1686886	12/2004	CN	N/A
S5584529	12/1979	JP	N/A
S5584547	12/1979	JP	N/A
S63200848	12/1987	JP	N/A
H02119952	12/1989	JP	N/A
H03174355	12/1990	JP	N/A
H05228386	12/1992	JP	N/A
H05253512	12/1992	JP	N/A
H05294610	12/1992	JP	N/A
H05329394	12/1992	JP	N/A

H0671193	12/1993	JP	N/A
H08116098	12/1995	JP	N/A
H08130142	12/1995	JP	N/A
H10144509	12/1997	JP	N/A
H11323508	12/1998	JP	N/A
2004031761	12/2003	JP	N/A
2004071854	12/2003	JP	N/A
2006066870	12/2005	JP	N/A
2006147830	12/2005	JP	N/A
2006238604	12/2005	JP	N/A
2008097677	12/2007	JP	N/A
2010267637	12/2009	JP	N/A
2011077223	12/2010	JP	N/A
2011211106	12/2010	JP	N/A
4887617	12/2011	JP	N/A
2013089813	12/2012	JP	N/A
2015195326	12/2014	JP	N/A
2016044352	12/2015	JP	N/A
2017171949	12/2016	JP	N/A
2018127716	12/2017	JP	N/A
2019012796	12/2018	JP	N/A
6489073	12/2018	JP	N/A
6555197	12/2018	JP	N/A
6614647	12/2018	JP	N/A
6759855	12/2019	JP	N/A
2021055133	12/2020	JP	N/A
2021077697	12/2020	JP	N/A
2005096325	12/2004	WO	N/A
2015199096	12/2014	WO	N/A
2021085521	12/2020	WO	N/A

OTHER PUBLICATIONS

Machine translation of JP 2004-071854A. (Year: 2004). cited by examiner

Machine translation of JP4887617B2. (Year: 2012). cited by examiner

Final Office Action issued in U.S. Appl. No. 17/806,300, dated Jul. 31, 2023, 23 pages. cited by applicant

Final Office Action issued in U.S. Appl. No. 17/935,907, dated Jul. 31, 2023, 20 pages. cited by applicant

United States Patent and Trademark office, Non-Final office action issued to U.S. Appl. No. 17/806,300 on Feb. 16, 2023, 20 pages. cited by applicant

United States Patent and Trademark office, Non-Final office action issued to U.S. Appl. No. 17/935,907 on Feb. 16, 2023, 17 pages. cited by applicant

Primary Examiner: Su; Xiaowei

Attorney, Agent or Firm: Hunton Andrews Kurth LLP

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2021-097489 filed on Jun. 10, 2021, and Japanese Patent Application No. 2022-088565 filed on May 31, 2022. The disclosures of Japanese Patent Application No. 2021-097489 and Japanese Patent Application No. 2022-088565 are hereby incorporated by reference in their entirety.

BACKGROUND

(2) The present disclosure relates to a method of producing a SmFeN-based anisotropic magnetic powder and a SmFeN-based anisotropic magnetic powder.

(3) JP 2015-195326 A discloses a production method involving grinding a SmFeN-based anisotropic magnetic powder using ceramic media in a solvent. However, the use of hard ceramic media is considered to cause chipping to form fine particles, so that the ground SmFeN-based anisotropic magnetic powder has a higher oxygen content and lower magnetic properties.

SUMMARY

(4) An exemplary object of the present disclosure is to provide a SmFeN-based anisotropic magnetic powder having excellent magnetic properties and a low oxygen content, and a method of producing the powder.

(5) Exemplary embodiments of the present disclosure relate to a method of producing a SmFeN-based anisotropic magnetic powder, the method including preparing a SmFeN-based anisotropic magnetic powder before dispersing comprising Sm, Fe, W, and N, and dispersing the SmFeN-based anisotropic magnetic powder before dispersing using a resin-coated metal media or a resin-coated ceramic media to obtain a SmFeN-based anisotropic magnetic powder.

(6) Exemplary embodiments of the present disclosure relate to a SmFeN-based anisotropic magnetic powder, comprising Sm, Fe, W, and N and having an average particle size of less than 2.5 μm , a residual magnetization σ_r of not less than 130 emu/g, and an oxygen content of not higher than 0.75% by mass.

(7) According to the above embodiments, it is possible to provide a SmFeN-based anisotropic magnetic powder having excellent magnetic properties and a low oxygen content, and a method of producing the powder.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 shows an exemplary SEM image of a magnetic powder prepared in Example 1.

(2) FIG. 2 shows an exemplary SEM image of a magnetic powder prepared in Comparative Example 1.

(3) FIG. 3 shows an exemplary SEM image of a magnetic powder prepared in Comparative Example 2.

(4) FIG. 4 shows an exemplary SEM image of a magnetic powder prepared in Example 2.

DETAILED DESCRIPTION

(5) Embodiments of the present disclosure are described in detail below. The following embodiments, however, are intended as examples to embody the technical idea of the present disclosure and are not intended to limit the scope of the present disclosure to the following embodiments. As used herein, the term “step” encompasses not only an independent step but also a step that may not be clearly distinguished from other steps, as long as a desired object of the step is achieved. Moreover, numerical ranges indicated using “to” refer to ranges including the numerical values indicated before and after “to” as the minimum and maximum, respectively.

(6) A method of producing a SmFeN-based anisotropic magnetic powder according to embodiments of the present disclosure includes preparing a SmFeN-based anisotropic magnetic

powder before dispersing comprising Sm, Fe, W, and N, and dispersing the SmFeN-based anisotropic magnetic powder before dispersing using a resin-coated metal media or a resin-coated ceramic media to obtain a SmFeN-based anisotropic magnetic powder.

(7) Dispersion Step

(8) In the dispersion step, a SmFeN-based anisotropic magnetic powder containing Sm, Fe, W, and N may be dispersed using a media such as a resin-coated metal media or a resin-coated ceramic media. Herein, the term “dispersion”, “dispersing”, or “dispersed” means that the aggregated particles in the SmFeN-based anisotropic magnetic powder formed by sintering or magnetic aggregation are separated into single particles or particles consisting of very few particles (hereinafter, referred to as single particles). According to the present embodiments, the inclusion of W in a process leading to the dispersion step allows the resulting SmFeN-based anisotropic magnetic powder to have a relatively small average particle size (for example, less than 2.5 μm). Further, since the impact energy of collision between the SmFeN-based anisotropic magnetic powder and the resin-coated metal media or the resin-coated ceramic media is smaller than that of collision between the SmFeN-based anisotropic magnetic powder and a non-resin coated metal media or a non-resin coated ceramic media, dispersion is more likely to occur than grinding. If the SmFeN-based anisotropic magnetic powder is ground as in the conventional art, the average particle size is greatly reduced, and fine particles are also formed due to chipping, likely resulting in a reduction in magnetic properties. In addition, since highly active new surfaces are generated on the fine particles and on the parts from which the fine particles are produced, oxidation is likely to occur, resulting in a higher oxygen content. In contrast, when dispersion is performed as in the present embodiments, it is considered that the formed single particles can be easily oriented in a magnetic field to enhance the magnetic properties; further, the formation of new surfaces associated with fine particle formation can be suppressed as compared to in grinding, so that the oxygen content is less likely to increase.

(9) The dispersion apparatus used in the dispersion step may be a vibration mill, for example. The media used in the dispersion apparatus such as the vibration mill may include a metal core and a coating resin coating the metal core. Examples of the material of the metal core include iron, chromium steel, stainless steel, and steel. The media used in the dispersion apparatus such as the vibration mill may include a ceramic core and a coating resin coating the ceramic core. Examples of the material of the ceramic core include inorganic compounds such as oxides, carbides, nitrides, or borides of metals or non-metals, specific examples of which include alumina, silica, zirconia, silicon carbide, silicon nitride, barium titanate, and glass. Iron or chromium steel is preferred among these because they have a high dispersing ability owing to the high specific gravity and less wear owing to the high hardness, and also because the iron-containing wear powder generated by abrasion has a low impact on the SmFeN-based anisotropic magnetic powder. Therefore, it is preferred that a media of a resin-coated iron core or a resin-coated chromium steel core is used in the dispersion apparatus. Examples of the coating resin include thermoplastic resins such as nylon 6, nylon 66, nylon 12, polypropylene, polyphenylene sulfide, and polyethylene, and thermosetting resins such as epoxy resins and silicone resins, and combinations thereof. Nylon such as nylon 6, nylon 66, nylon 12 is preferred among these. A media of a nylon-coated iron core may be used in the dispersion apparatus.

(10) The media used in the dispersion step preferably has a specific gravity of not less than 4, more preferably not less than 5. When the specific gravity is less than 4, the impact energy during dispersion tends to be too small so that dispersion is less likely to occur. The upper limit of the specific gravity is not limited, but is preferably not more than 8, more preferably not more than 7.5. The media used in the dispersion step may have a specific gravity of at least 6 but not more than 7.5. The media may include a core of a metal or a ceramic and a resin film coating the core.

(11) Although the dispersion step may be performed in the presence of a solvent, it is preferably performed in the absence of a solvent in order to suppress the oxidation of the SmFeN-based

anisotropic magnetic powder by the components (e.g., moisture) in the solvent.

(12) To suppress oxidation of the SmFeN-based anisotropic magnetic powder, the dispersion step is preferably performed in an inert gas atmosphere such as a nitrogen gas atmosphere or an argon gas atmosphere. The concentration of nitrogen in the nitrogen gas atmosphere may be 90% by volume or more, and preferably 95% by volume or more. The concentration of argon in the argon gas atmosphere may be 90% by volume or more, and preferably 95% by volume or more. The inert gas atmosphere may be an atmosphere in which two or more inert gases such as nitrogen gas and argon gas are mixed. The concentration of the inert gas in the inert gas atmosphere may be 90% by volume or more, and preferably 95% by volume or more.

(13) The diameter of the resin-coated metal media or the resin-coated ceramic media is preferably at least 2 mm and not more than 100 mm, more preferably at least 3 mm and not more than 15 mm, still more preferably at least 3 mm and not more than 10 mm. The media having a diameter of less than 2 mm is difficult to be coated with the resin, while the media having a diameter of more than 100 mm is large and thus tends to have less contact with the powder so that dispersion is less likely to occur.

(14) When a vibration mill is used in the dispersion step, for example, the amount of the media may be at least 60% by volume but not more than 70% by volume, and the amount of the SmFeN-based anisotropic magnetic powder may be at least 3% by volume but not more than 20% by volume, preferably at least 5% by volume but not more than 20% by volume, each relative to the volume of the container used to contain the SmFeN-based anisotropic magnetic powder and the media.

(15) Although the SmFeN-based anisotropic magnetic powder before dispersing used in the dispersion step may be produced with reference to the method disclosed in, for example, JP 2017-117937 A or JP 2021-055188 A, an exemplary method of producing the SmFeN-based anisotropic magnetic powder before dispersing will be described below. The SmFeN-based anisotropic magnetic powder before dispersing is a magnetic powder before performing the dispersion step described above. The SmFeN-based anisotropic magnetic powder before dispersing may have been pre-dispersed by a different step than the dispersion step described above.

(16) The SmFeN-based anisotropic magnetic powder before dispersing used in the dispersion step may be prepared by a production method including: pretreating an oxide containing Sm, Fe, and W by heat treatment in a reducing gas-containing atmosphere to obtain a partial oxide; heat treating the partial oxide in the presence of a reducing agent to obtain alloy particles; nitriding the alloy particles to obtain a nitride; and washing the nitride to obtain the SmFeN-based anisotropic magnetic powder before dispersing.

(17) Although the oxide containing Sm, Fe, and W used in the pretreatment step may be prepared by mixing a Sm oxide, a Fe oxide, and a W oxide, it can be prepared by mixing a solution containing Sm, Fe, and W with a precipitating agent to obtain a precipitate containing Sm, Fe, and W (precipitation step), and calcining the precipitate to obtain an oxide containing Sm, Fe, and W (oxidation step).

(18) Precipitation Step

(19) In the precipitation step, a Sm source, a Fe source, and a W source may be dissolved to prepare a solution containing Sm, Fe, and W. When the main phase to be obtained is Sm₂Fe₁₇N₃, the molar ratio of Sm and Fe (Sm:Fe) is preferably 1.5:17 to 3.0:17, more preferably 2.0:17 to 2.5:17. To the aforementioned solution may be added La, Co, Ti, Sc, Y, Pr, Nd, Pm, Gd, Tb, Dy, Ho, Er, Tm, Lu, or other sources, in addition to W. In view of remanence, the solution preferably contains La. In view of temperature characteristics, the solution preferably contains Co or Ti.

(20) Any soluble Sm, Fe, or W source may be used. In view of availability, for example, the Sm source may be samarium oxide, the Fe source may be FeSO₄, and the W source may be ammonium tungstate. The concentration of the solution containing Sm, Fe, and W may be appropriately adjusted within a range in which the Sm, Fe, and W sources can be substantially

dissolved.

(21) The solution containing Sm, Fe, and W may be reacted with a precipitating agent to obtain an insoluble precipitate containing Sm, Fe, and W. Here, the solution containing Sm, Fe, and W is not limited as long as Sm, Fe, and W are present in the solution during the reaction with the precipitating agent. For example, a solution containing Sm, a solution containing Fe, and a solution containing W may be prepared as separate solutions and individually added dropwise to react with the precipitating agent. Alternatively, the solution containing Sm, Fe, and W may be such that a solution containing Sm and Fe and a solution containing W may be prepared as separate solutions and individually added dropwise to react with the precipitating agent. When separate solutions are prepared, the concentration of each solution may also be appropriately adjusted within a range in which the corresponding source(s) can be substantially dissolved. The precipitating agent may be any alkaline solution that reacts with the solution containing Sm, Fe, and W to give a precipitate. Examples include ammonia water and caustic soda, with caustic soda being preferred.

(22) In order to easily control the particle properties of the precipitate, the precipitation reaction is preferably performed by adding dropwise the solution containing Sm, Fe, and W and the precipitating agent each to a solvent such as water. A precipitate having a homogeneous element distribution, a narrow particle size distribution, and a uniform particle shape can be obtained by appropriately controlling the feeding rates of the solution containing Sm, Fe, and W and the precipitating agent, the reaction temperature, the concentration of the reaction solution, the pH during the reaction, and other conditions. The use of such a precipitate improves the magnetic properties of the finally produced SmFeN-based anisotropic magnetic powder. The reaction temperature is preferably at least 0° C. but not higher than 50° C., more preferably at least 35° C. but not higher than 45° C. The concentration of the reaction solution calculated as the total concentration of metal ions is preferably at least 0.65 mol/L but not more than 0.85 mol/L, more preferably at least 0.7 mol/L but not more than 0.85 mol/L. The reaction pH is preferably at least 5 but not more than 9, more preferably at least 6.5 but not more than 8.

(23) In view of magnetic properties, the solution containing Sm, Fe, and W preferably further contains at least one metal selected from the group consisting of La, Co, and Ti. For example, in view of remanence, the solution preferably contains La, while in view of temperature characteristics the solution preferably contains Co or Ti. The La source is not limited as long as it is soluble in a strongly acidic solution. In view of availability, examples include La.sub.2O.sub.3 and LaCl.sub.3. The concentration may be appropriately adjusted within a range in which the Sm, Fe, and W sources and the La, Co, and/or Ti source can be dissolved in the solution. The Co source may be cobalt sulfate, and the titanium source may be sulfated titania.

(24) When the solution containing Sm, Fe, and W further contains at least one metal selected from the group consisting of La, Co, and Ti, an insoluble precipitate containing Sm, Fe, W, and at least one selected from the group consisting of La, Co, and Ti will be produced. Here, the solution is not limited as long as at least one selected from the group consisting of La, Co, and Ti is present in the solution during the reaction with the precipitating agent. For example, the sources may be prepared as separate solutions and individually added dropwise to react with the precipitating agent. Alternatively, they may be prepared into the same solution containing Sm, Fe, and W.

(25) The powder obtained in the precipitation step roughly determines the powder particle size, particle shape, and particle size distribution of the finally produced SmFeN-based anisotropic magnetic powder. When the particle size of the obtained powder is measured with a laser diffraction-type wet particle size distribution analyzer, the size and distribution of all the powder preferably substantially fall within the range of at least 0.05 μm but not more than 20 μm , preferably at least 0.1 μm but not more than 10 μm .

(26) After separating the precipitate, the separated precipitate is preferably subjected to solvent removal in order to reduce aggregation of the precipitate caused by evaporation of the residual solvent in which the precipitate has been re-dissolved during the heat treatment in the subsequent

oxidation step, and to reduce changes in properties such as particle size distribution and powder particle size. Specifically, when the solvent used is water, for example, the solvent removal may be performed by drying in an oven at at least 70° C. but not higher than 200° C. for at least 5 hours but not longer than 12 hours.

(27) The precipitation step may be followed by washing and separating the resulting precipitate. The step of washing may be appropriately performed until the conductivity of the supernatant solution reaches 5 mS/m.^{sup.2} or lower. The step of separating the precipitate may be performed, for example, by mixing the resulting precipitate with a solvent (preferably water), followed by filtration, decantation, or other separation methods.

(28) Oxidation Step

(29) The oxidation step includes calcining the precipitate formed in the precipitation step to obtain an oxide containing Sm, Fe, and W. For example, the precipitate may be converted into an oxide by heat treatment. The heat treatment of the precipitate needs to be performed in the presence of oxygen, for example in an air atmosphere. Moreover, since the presence of oxygen is necessary, the non-metal portions of the precipitate preferably contain oxygen atoms.

(30) The heat treatment temperature in the oxidation step (hereinafter, oxidation temperature) is not limited, but is preferably at least 700° C. but not higher than 1,300° C., more preferably at least 900° C. but not higher than 1,200° C. When the temperature is lower than 700° C., the oxidation tends to be insufficient. When the temperature is higher than 1,300° C., the resulting SmFeN-based anisotropic magnetic powder tends not to have the target particle shape, average particle size, and particle size distribution. The heat treatment duration is not limited either, but is preferably at least 1 hour but not longer than 3 hours.

(31) The thus formed oxide is oxide particles in which Sm and Fe have been sufficiently microscopically mixed, and the particle shape, particle size distribution, and other properties of the precipitate have been reflected.

(32) Pretreatment Step

(33) The pretreatment step includes subjecting the oxide containing Sm, Fe, and W to heat treatment in a reducing gas-containing atmosphere to obtain a partial oxide which is a partially reduced product of the oxide.

(34) Here, the term “partial oxide” refers to a partially reduced oxide. The oxygen concentration of the partial oxide is not limited, but is preferably not more than 10% by mass, more preferably not more than 8% by mass. When the concentration is more than 10% by mass, the heat generated by reduction with Ca in the reduction step tends to increase, raising the calcination temperature enough to form abnormally grown particles. Here, the oxygen concentration of the partial oxide can be measured by a non-dispersive infrared spectroscopy (ND-IR).

(35) The reducing gas may be appropriately selected from, for example, hydrogen (H.^{sub.2}), carbon monoxide (CO), hydrocarbon gases such as methane (CH.^{sub.4}), and combinations thereof. Hydrogen gas is preferred in terms of cost. The flow rate of the gas may be appropriately adjusted within a range that does not cause scattering of the oxide. The heat treatment temperature in the pretreatment step (hereinafter, pretreatment temperature) is preferably at least 300° C. but not higher than 950° C. The lower limit is more preferably at least 400° C., still more preferably at least 750° C. The upper limit is more preferably lower than 900° C. When the pretreatment temperature is at least 300° C., the oxide containing Sm and Fe can be efficiently reduced. When the pretreatment temperature is not higher than 950° C., the grain growth and segregation of the oxide particles can be inhibited so that the desired particle size can be maintained. The heat treatment duration is not limited, but may be at least 1 hour but not longer than 50 hours. Moreover, when the reducing gas used is hydrogen, preferably the thickness of the oxide layer used is adjusted to not more than 20 mm, and further the dew point in the reaction furnace is adjusted to not higher than -10° C.

(36) Reduction Step

(37) The reduction step includes heat treating the partial oxide in the presence of a reducing agent to obtain alloy particles. For example, the reduction may be performed by contacting the partial oxide with molten calcium or calcium vapor. In view of magnetic properties, the heat treatment temperature is preferably at least 920° C. but not higher than 1,200° C., more preferably at least 950° C. but not higher than 1,150° C., still more preferably at least 960° C. but not higher than 1,000° C.

(38) As an alternative to the above-mentioned heat treatment process in the reduction step, heat treatment may be performed at a first temperature of at least 950° C. but not higher than 1,030° C., and then at a second temperature lower than the first temperature of at least 930° C. but not higher than 1,000° C. The first temperature is preferably at least 960° C. but not higher than 1,000° C., and the second temperature is preferably at least 940° C. but not higher than 980° C. With regard to the difference between the first temperature and the second temperature, the second temperature is preferably lower than the first temperature by at least 10° C. but not more than 60° C., more preferably by at least 10° C. but not more than 30° C. The heat treatment at the first temperature and the heat treatment at the second temperature may be continuously performed. Although there may be a heat treatment at a temperature lower than the second temperature between these heat treatments, it is preferred in view of productivity to perform these treatments continuously. To perform a more uniform reduction reaction, the duration of each heat treatment is preferably shorter than 120 minutes, more preferably shorter than 90 minutes. The lower limit of the heat treatment duration is preferably not shorter than 10 minutes, more preferably not shorter than 30 minutes.

(39) The metallic calcium serving as a reducing agent may be used in the form of granules or powder, and its average particle size is preferably 10 mm or less in order to more effectively reduce aggregation during the reduction reaction. Moreover, the metallic calcium is preferably added in an amount that is 1.1 to 3.0 times, more preferably 1.5 to 2.5 times the reaction equivalent (which is the stoichiometric amount needed to reduce the rare earth oxides, but includes the amount needed to reduce oxides of the Fe component, if present).

(40) In the reduction step, the metallic calcium as a reducing agent may be used in combination with a disintegration accelerator, if necessary. The disintegration accelerator may be appropriately used to facilitate the disintegration or granulation of the product during the post treatment step described later. Examples include alkaline earth metal salts such as calcium chloride, and alkaline earth oxides such as calcium oxide. Such a disintegration accelerator may be used in an amount of at least 1% by mass but not more than 30% by mass, preferably at least 5% by mass but not more than 30% by mass, relative to the amount of the samarium oxide.

(41) Nitridation Step

(42) The nitridation step includes nitriding the alloy particles obtained in the reduction step to obtain anisotropic magnetic particles. Since the particulate precipitate obtained in the precipitation step is used, the alloy particles obtained in the reduction step are in porous bulk form. This permits the alloy particles to be directly nitrided by heat treatment in a nitrogen atmosphere without grinding, resulting in uniform nitridation.

(43) The heat treatment temperature in the nitridation of the alloy particles (hereinafter, nitridation temperature) is preferably adjusted at 300 to 610° C., particularly preferably 400 to 550° C., and the atmosphere may be replaced with nitrogen to perform the heat treatment in this temperature range. The heat treatment duration may be set so that the alloy particles can be sufficiently uniformly nitrided.

(44) With regard to the heat treatment temperature in the nitridation of the alloy particles, heat treatment may be performed at a first temperature of at least 400° C. but not higher than 470° C. and then at a second temperature of at least 480° C. but not higher than 610° C. for nitridation. If the alloy particles are heat treated at the high second temperature without being nitrided at the first temperature, the nitridation may rapidly proceed to cause abnormal heat generation which can degrade the SmFeN-based anisotropic magnetic powder, greatly reducing the magnetic properties.

Moreover, the nitridation step is preferably performed in a substantially nitrogen atmosphere in order to allow the nitridation to proceed more slowly.

(45) Here, the term “substantially” is used in consideration of the potential presence of unavoidable element(s) other than nitrogen due to contamination of impurities or other factors. For example, the nitrogen content of the atmosphere is not lower than 95%, preferably not lower than 97%, more preferably not lower than 99%.

(46) The first temperature in the nitridation step is preferably at least 400° C. but not higher than 470° C., more preferably at least 410° C. but not higher than 450° C. When the first temperature is lower than 400° C., the nitridation tends to proceed very slowly. When the first temperature is higher than 470° C., excessive nitridation or degradation tends to easily occur due to heat generation. The heat treatment duration at the first temperature is not limited, but is preferably at least 1 hour but not longer than 40 hours, more preferably not longer than 20 hours. When the duration is shorter than 1 hour, the nitridation may insufficiently proceed. When the duration is longer than 40 hours, productivity is impaired.

(47) The second temperature is preferably at least 480° C. but not higher than 610° C., more preferably at least 500° C. but not higher than 550° C. When the second temperature is lower than 480° C., the nitridation of large particles may insufficiently proceed. When the second temperature is higher than 610° C., excessive nitridation or degradation can easily occur. The heat treatment duration at the second temperature is preferably at least 15 minutes but not longer than 5 hours, more preferably at least 30 minutes but not longer than 2 hours. When the duration is shorter than 15 minutes, the nitridation may insufficiently proceed. When the duration is longer than 5 hours, productivity is impaired.

(48) The heat treatment at the first temperature and the heat treatment at the second temperature may be continuously performed. Although there may be a heat treatment at a temperature lower than the second temperature between these heat treatments, it is preferred in view of productivity to perform these treatments continuously.

(49) Post Treatment Step

(50) In some cases, the product obtained after the nitridation step contains, in addition to the magnetic particles, contaminants such as by-product CaO and unreacted metallic calcium, and forms a composite with these contaminants in sintered bulk form. Such a product obtained after the nitridation step may be introduced into cold water to separate the CaO and metallic calcium as a suspension of calcium hydroxide (Ca(OH)₂) from the SmFeN-based anisotropic magnetic powder. Further, the residual calcium hydroxide may be sufficiently removed by washing the SmFeN-based anisotropic magnetic powder with acetic acid or the like. When the product is introduced into water, oxidation of metallic calcium by water and hydration of by-product CaO will occur, causing disintegration or micronization of the reaction product that is a composite in sintered bulk form.

(51) Alkali Treatment Step

(52) The product obtained after the nitridation step may be introduced into an alkali solution. Examples of the alkali solution used in the alkali treatment step include an aqueous calcium hydroxide solution, an aqueous sodium hydroxide solution, and an aqueous ammonia solution. In view of waste water treatment and high pH, an aqueous calcium hydroxide solution or an aqueous sodium hydroxide solution is preferred among these. In the alkali treatment of the product obtained after the nitridation step, the remaining Sm-rich layer containing a certain amount of oxygen serves as a protection layer, thereby reducing an increase in oxygen concentration caused by the alkali treatment.

(53) The pH of the alkali solution used in the alkali treatment step is not limited, but is preferably not less than 9, more preferably not less than 10. When the pH is less than 9, the rate of the reaction into calcium hydroxide is high, causing more heat generation. Thus, the finally produced SmFeN-based anisotropic magnetic powder tends to have a higher oxygen concentration.

(54) In the alkali treatment step, the SmFeN-based anisotropic magnetic powder obtained after the treatment with an alkali solution may optionally be subjected to decantation or other techniques to remove the moisture.

(55) Acid Treatment Step

(56) The alkali treatment step may further be followed by treatment with an acid. In the acid treatment step, the aforementioned Sm-rich layer may be at least partially removed to reduce the oxygen concentration of the magnetic powder as a whole. Moreover, since the production method according to embodiments of the present disclosure does not include grinding or the like, the SmFeN-based anisotropic magnetic powder has a small average particle size and a narrow particle size distribution, and also does not contain fine particles formed by grinding or the like, which makes it possible to reduce an increase in oxygen concentration.

(57) Any acid may be used in the acid treatment step. Examples include hydrogen chloride, nitric acid, sulfuric acid, and acetic acid. To avoid residual impurities, hydrogen chloride or nitric acid is preferred among these.

(58) The amount of the acid used in the acid treatment step per 100 parts by mass of the SmFeN-based anisotropic magnetic powder is preferably at least 3.5 parts by mass but not more than 13.5 parts by mass, more preferably at least 4 parts by mass but not more than 10 parts by mass. When the amount is less than 3.5 parts by mass, the oxide tends to remain on the surface of the SmFeN-based anisotropic magnetic powder, resulting in a higher oxygen concentration. When the amount is more than 13.5 parts by mass, reoxidation is more likely to occur upon exposure to the air, and the cost also tends to increase because the acid dissolves the SmFeN-based anisotropic magnetic powder. When the amount of the acid is at least 3.5 parts by mass but not more than 13.5 parts by mass per 100 parts by mass of the SmFeN-based anisotropic magnetic powder, the surface of the SmFeN-based anisotropic magnetic powder can be coated with a Sm-rich layer which is oxidized enough to inhibit reoxidation upon exposure to the air after the acid treatment. Thus, the resulting SmFeN-based anisotropic magnetic powder has a low oxygen concentration, a small average particle size, and a narrow particle size distribution.

(59) In the acid treatment step, the SmFeN-based anisotropic magnetic powder obtained after the treatment with an acid may optionally be subjected to decantation or other techniques to reduce the moisture.

(60) Dehydration Step

(61) The acid treatment step is preferably followed by dehydration. The dehydration can reduce the moisture in the solids before vacuum drying, thereby inhibiting the progress of oxidation during drying caused due to the higher moisture content of the solids before vacuum drying. Here, the term “dehydration” refers to a treatment in which a pressure or a centrifugal force is applied to reduce the moisture content of the solids after the treatment as compared to that of the solids before the treatment, and excludes mere decantation, filtration, or drying. The dehydration may be performed by any method such as squeezing or centrifugation.

(62) The moisture content of the SmFeN-based anisotropic magnetic powder after the dehydration is not limited, but in order to inhibit the progress of oxidation, it is preferably not higher than 13% by mass, more preferably not higher than 10% by mass.

(63) The SmFeN-based anisotropic magnetic powder obtained by acid treatment or the SmFeN-based anisotropic magnetic powder obtained by acid treatment followed by dehydration is preferably dried in vacuum. The drying temperature is not limited, but is preferably not lower than 70° C., more preferably not lower than 75° C. The drying duration is not limited either, but is preferably not shorter than 1 hour, more preferably not shorter than 3 hours.

(64) Surface Treatment Step

(65) The SmFeN-based anisotropic magnetic powder obtained in the post treatment step may be subjected to surface treatment. For example, a phosphoric acid solution as a surface treatment agent may be introduced in the range of 0.10% to 10% by mass, as calculated as $\text{PO}_{0.4}$, relative to the

solids content of the magnetic powder obtained in the nitridation step. The magnetic powder may be appropriately separated from the solution and dried to obtain a surface-treated SmFeN-based anisotropic magnetic powder.

(66) The SmFeN-based anisotropic magnetic powder according to certain exemplary embodiments of the present disclosure contains Sm, Fe, W, and N and has an average particle size of less than 2.5 μm , a residual magnetization σ_r of not less than 130 emu/g, and an oxygen content of not higher than 0.75% by mass.

(67) The average particle size of the SmFeN-based anisotropic magnetic powder may be, for example, less than 2.5 μm , or at least 0.5 μm but not more than 2.4 μm , preferably at least 1.0 μm but not more than 2.0 μm . Here, the term “average particle size” refers to the particle size measured using a laser diffraction particle size distribution analyzer under a dry condition.

(68) The particle size D10 of the SmFeN-based anisotropic magnetic powder is preferably not less than 0.3 μm , more preferably not less than 0.5 μm . When the D10 is less than 0.3 μm , the magnetization of the SmFeN-based anisotropic magnetic powder tends to greatly decrease. Here, the term “D10” refers to the particle size corresponding to the 10th percentile of the cumulative particle size distribution by volume of the SmFeN-based anisotropic magnetic powder.

(69) The particle size D50 of the SmFeN-based anisotropic magnetic powder is preferably at least 0.5 μm but not more than 2.5 μm , more preferably at least 1.0 μm but not more than 2.0 μm . When the D50 is less than 0.5 μm , the amount of the SmFeN-based anisotropic magnetic powder filled in the bonded magnet tends to decrease, resulting in lower magnetization. When the D50 is more than 2.0 μm , the magnetic powder tends to aggregate, resulting in lower magnetic properties. Here, the term “D50” refers to the particle size corresponding to the 50th percentile of the cumulative particle size distribution by volume of the SmFeN-based anisotropic magnetic powder.

(70) The particle size D90 of the SmFeN-based anisotropic magnetic powder is preferably at least 2 μm but not more than 5 μm , more preferably at least 2.5 μm but not more than 3.5 μm . When the D90 is less than 2 μm , the amount of the SmFeN-based anisotropic magnetic powder filled in the bonded magnet tends to decrease, resulting in lower magnetization. When the D90 is more than 3.5 μm , the coercive force of the bonded magnet tends to decrease. Here, the term “D90” refers to the particle size corresponding to the 90th percentile of the cumulative particle size distribution by volume of the SmFeN-based anisotropic magnetic powder.

(71) The residual magnetization σ_r is not less than 130 emu/g, preferably not less than 131 emu/g.

(72) The oxygen content of the SmFeN-based anisotropic magnetic powder is not higher than 0.75% by mass, preferably not higher than 0.65% by mass, more preferably not higher than 0.6% by mass. When the oxygen content is higher than 0.75% by mass, a lot of oxygen can be present on the particle surface, causing the formation of $\alpha\text{-Fe}$. Here, the oxygen content is analyzed after the SmFeN-based anisotropic magnetic powder obtained after completion of all the steps is allowed to stand in the air for at least 30 minutes.

(73) The SmFeN-based anisotropic magnetic powder according to the present embodiments is typically represented by the following formula:

$\text{Sm}_{\text{sub.v}}\text{Fe}_{\text{sub.}(100-\text{v}-\text{w}-\text{x}-\text{y}-\text{z}-\text{u})}\text{N}_{\text{sub.w}}\text{La}_{\text{sub.x}}\text{W}_{\text{sub.y}}\text{Co}_{\text{sub.z}}\text{Ti}_{\text{sub.u}}$

wherein $3 \leq v \leq 30$, $5 \leq w \leq 15$, $0 \leq x \leq 0.3$, $0 \leq y \leq 2.5$, $0 \leq z \leq 2.5$, and $0 \leq u \leq 2.5$.

(74) In the formula, v is defined to be at least 3 but not more than 30 for the following reason. If v is less than 3, the unreacted iron component ($\alpha\text{-Fe}$ phase) may be separated, which reduces the coercive force of the SmFeN-based anisotropic magnetic powder so as to fail to provide a practical magnet, while if v is more than 30, the Sm element may precipitate and make the SmFeN-based anisotropic magnetic powder unstable in the air, thereby reducing the remanence. Moreover, w is defined to be at least 5 but not more than 15 for the following reason. If w is less than 5, almost no coercive force may be obtained, while if w is more than 15, a nitride of Sm or iron itself may be formed. Moreover, y is defined to be more than 0 but not more than 2.5 for the following reason. If y is more than 2.5, a nitride of Sm or iron itself may be formed, thereby greatly reducing the

magnetization.

(75) In view of remanence, the amount of La, if present, is preferably at least 0.1% by mass but not more than 5% by mass, more preferably at least 0.15% by mass but not more than 1% by mass.

(76) In view of temperature characteristics, the amount of Co, if present, is preferably at least 0.1% by mass but not more than 5% by mass, more preferably at least 0.15% by mass but not more than 1% by mass.

(77) In view of temperature characteristics, the amount of Ti, if present, is preferably at least 0.1% by mass but not more than 5% by mass, more preferably at least 0.15% by mass but not more than 1% by mass.

(78) The amount of N is preferably at least 3.3% by mass but not more than 3.5% by mass. When the amount is more than 3.5% by mass, excessive nitridation may occur. When the amount is less than 3.3% by mass, insufficient nitridation may occur. In both cases, the magnetic properties tend to decrease.

(79) In particular, SmFeWN or SmFeWLaN is preferred.

(80) The SmFeN-based anisotropic magnetic powder may have a below-defined span of not more than 2, preferably not more than 1.8, still more preferably not more than 1.6.

$\text{Span} = (D90 - D10) / D50$

(81) In the formula, D10, D50, and D90 represent the particle sizes corresponding to the 10th percentile, 50th percentile, and 90th percentile, respectively, of the cumulative particle size distribution by volume. When the span is more than 2, large particles are present, so that the magnetic properties tend to decrease.

(82) The average circularity of the SmFeN-based anisotropic magnetic powder is preferably not less than 0.50, more preferably not less than 0.70, particularly preferably not less than 0.75. When the circularity is less than 0.50, the fluidity may deteriorate so that stress can occur between the particles during magnetic field compaction, resulting in lower magnetic properties. The circularity may be determined using a scanning electron microscope (SEM) and a particle analysis Ver. 3 available from Sumitomo Metal Technology, Inc. as image analysis software. The circularity may be determined by taking a SEM image at a magnification of 3,000, processing the image for binarization, and calculating the circularity of each particle. The term “circularity” defined in the present disclosure refers to the average of the circularities obtained by measuring about 1,000 to 10,000 particles. In general, the larger the number of small size particles, the higher the circularity. Hence, particles having a particle size of not less than 1 μm are measured for circularity. The circularity measurement uses the definitional equation: $\text{Circularity} = 4 \pi S / L$, wherein S represents the area of the two-dimensional projection of the particle, and L represents the perimeter of the two-dimensional projection thereof.

(83) The SmFeN-based anisotropic magnetic powder according to the present embodiments has high residual magnetization and thus is usable as a sintered magnet or a bonded magnet, for example.

(84) A bonded magnet may be produced from the SmFeN-based anisotropic magnetic powder according to the present embodiments and a resin. The inclusion of the SmFeN-based anisotropic magnetic powder enables production of a composite material having high magnetic properties.

(85) The resin contained in the composite material may be either a thermosetting resin or a thermoplastic resin, preferably a thermoplastic resin. Specific examples of the thermoplastic resin include polyphenylene sulfide resins (PPS), polyether ether ketones (PEEK), liquid crystal polymers (LCP), polyamides (PA), polypropylenes (PP), and polyethylenes (PE).

(86) The mass ratio of the resin to the SmFeN-based anisotropic magnetic powder (resin/SmFeN-based anisotropic magnetic powder) in the production of the composite material is preferably 0.10 to 0.15, more preferably 0.11 to 0.14.

(87) For example, the composite material may be obtained by mixing the SmFeN-based anisotropic magnetic powder and the resin using a kneader at 280 to 330° C.

(88) The composite material may be used to produce a bonded magnet. Specifically, for example, a bonded magnet may be produced by heat treating the composite material to align the easy axes of magnetization in an orientation field (orientation step), followed by pulse magnetization in a magnetizing field (magnetization step).

(89) The heat treatment temperature in the orientation step is preferably, for example, 90 to 200° C., more preferably 100 to 150° C. The magnitude of the orientation field in the orientation step may be, for example, 720 kA/m, while the magnitude of the magnetizing field in the magnetization step may be, for example, 1500 to 2500 kA/m.

(90) A sintered magnet may be produced by compacting and sintering the SmFeN-based anisotropic magnetic powder according to the present embodiments. The SmFeN-based anisotropic magnetic powder according to the present embodiments, which has a low oxygen concentration, a small average particle size, a narrow particle size distribution, and a high remanence, is suitable for sintered magnets.

(91) For example, the sintered magnet may be produced by sintering the SmFeN-based anisotropic magnetic powder in an atmosphere with an oxygen concentration of not more than 0.5 ppm by volume at a temperature of higher than 300° C. but lower than 600° C. under a pressure of at least 1,000 MPa but not more than 1,500 MPa, as described in JP 2017-055072A.

(92) For example, the sintered magnet may be produced by pre-compacting the SmFeN-based anisotropic magnetic powder in a magnetic field of not lower than 6 kOe, followed by warm compaction at a temperature of not higher than 600° C. and a contact pressure of 1 to 5 GPa, as described in WO2015/199096.

(93) For example, the sintered magnet may be produced by subjecting a mixture containing the SmFeN-based anisotropic magnetic powder and a metal binder to cold compaction at a contact pressure of 1 to 5 GPa, followed by heating at a temperature of 350 to 600° C. for 1 to 120 minutes, as described in JP 2016-082175 A.

EXAMPLES

(94) Examples are described below. It should be noted that “%” is by mass unless otherwise specified.

(95) Evaluation

(96) The metal contents, average particle size, particle size distribution, nitrogen content, oxygen content, and residual magnetization σ_r of the SmFeN-based anisotropic magnetic powder were evaluated as described below.

(97) Metal Contents

(98) The amounts of the metals (Sm, Fe, W, etc.) in the SmFeN-based anisotropic magnetic powder dissolved with hydrochloric acid were measured by ICP-AES (apparatus name: Optima 8300).

(99) Average Particle Size and Particle Size Distribution

(100) The average particle size and particle size distribution of the SmFeN-based anisotropic magnetic powder were measured with a laser diffraction particle size distribution analyzer (HELOS & RODOS available from Japan Laser Corporation).

(101) Nitrogen Content and Oxygen Content

(102) The nitrogen content and oxygen content of the SmFeN-based anisotropic magnetic powder were measured by a heat conductivity method (EMGA-820 available from Horiba Ltd.).

(103) Residual Magnetization σ_r , Coercive Force iH_c , and Squareness Ratio H_k

(104) The prepared SmFeN-based anisotropic magnetic powder was packed together with a paraffin wax into a sample vessel. After the paraffin wax was melted using a dryer, the easy axes of magnetization were aligned in an orientation field of 16 kA/m. The magnetically oriented sample was pulse magnetized in a magnetizing field of 32 kA/m, and the residual magnetization σ_r , coercive force iH_c , and squareness ratio H_k of the sample were measured using a vibrating sample magnetometer (VSM) with a maximum field of 16 kA/m.

Production Example 1

(105) Precipitation Step

(106) An amount of 5.0 kg of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ was mixed and dissolved in 2.0 kg of pure water. To the mixture were further added 0.49 kg of Sm_2O_3 and 0.74 kg of 70% sulfuric acid, and they were well stirred and completely dissolved. Next, pure water was added to the resulting solution so that the final Fe and Sm concentrations were adjusted to 0.726 mol/L and 0.112 mol/L, respectively, to obtain a SmFe sulfuric acid solution.

(107) The entire amount of the prepared SmFe sulfuric acid solution was added dropwise to 20 kg of pure water kept at a temperature of 40° C. with stirring over 70 minutes from the start of the reaction, while simultaneously adding dropwise 0.190 kg of a 13% by mass ammonium tungstate solution and a 15% by mass ammonia solution to adjust the pH to 7 to 8. Thus, a slurry containing a SmFeW hydroxide was obtained. The slurry was washed with pure water by decantation, followed by solid-liquid separation to separate the hydroxide. The separated hydroxide was dried in an oven at 100° C. for 10 hours.

(108) Oxidation Step

(109) The hydroxide obtained in the precipitation step was calcined in the air at 1,000° C. for 1 hour. After cooling, a red SmFeW oxide as raw material powder was obtained.

(110) Pretreatment Step

(111) An amount of 100 g of the SmFeW oxide obtained in Production Example 1 was put in a steel container to a thickness of 10 mm. The container was placed in a furnace, and the pressure was reduced to 100 Pa. Then, while introducing hydrogen gas, the temperature was increased to a pretreatment temperature of 850° C. and maintained for 15 hours. The oxygen concentration was measured by a non-dispersive infrared (ND-IR) analysis (EMGA-820 available from Horiba Ltd.) and found to be 5% by mass. The results show that a black partial oxide was obtained in which the oxygen bonded to Sm remained unreduced while 95% of the oxygen bonded to Fe was reduced.

(112) Reduction Step

(113) An amount of 60 g of the partial oxide obtained in the pretreatment step was mixed with 19.2 g of metallic calcium having an average particle size of about 6 mm, and the mixture was placed in a furnace. After vacuum evacuation of the furnace, argon gas (Ar gas) was introduced. The temperature was increased to a first temperature of 980° C. and maintained for 45 minutes to obtain SmFeW alloy particles.

(114) Nitridation Step

(115) Subsequently, the temperature inside the furnace was lowered to 100° C., followed by vacuum evacuation. Then, while introducing nitrogen gas, the temperature was increased to a first temperature of 430° C. and maintained for 3 hours. Next, the temperature was increased to a second temperature of 500° C. and maintained for 1 hour, followed by cooling to obtain a magnetic particle-containing bulk product.

(116) Post Treatment Step

(117) The bulk product obtained in the nitridation step was introduced into 3 kg of pure water and stirred for 30 minutes. After standing still, the supernatant was drained by decantation. The introduction into pure water, stirring, and decantation were repeated 10 times. Next, 2.5 g of 99.9% acetic acid was introduced and stirred for 15 minutes. After standing still, the supernatant was drained by decantation. The introduction into pure water, stirring, and decantation were repeated twice, followed by solid-liquid separation and then vacuum drying at 80° C. for 3 hours to obtain a SmFeN-based anisotropic magnetic powder.

Production Example 2

(118) Precipitation Step

(119) An amount of 5.0 kg of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ was mixed and dissolved in 2.0 kg of pure water. To the mixture were further added 0.49 kg of Sm_2O_3 , 0.035 kg of La_2O_3 , and 0.74 kg of 70% sulfuric acid, and they were well stirred and completely dissolved. Next, pure water was added to the resulting solution so that the final Fe and Sm concentrations were adjusted

to 0.726 mol/L and 0.112 mol/L, respectively, to obtain a SmFeLa sulfuric acid solution.

(120) The entire amount of the prepared SmFeLa sulfuric acid solution was added dropwise to 20 kg of pure water kept at a temperature of 40° C. with stirring over 70 minutes from the start of the reaction, while simultaneously adding dropwise 0.190 kg of a 13% by mass ammonium tungstate solution and a 15% by mass ammonia solution to adjust the pH to 7 to 8. Thus, a slurry containing a SmFeLaW hydroxide was obtained. The slurry was washed with pure water by decantation, followed by solid-liquid separation to separate the hydroxide. The separated hydroxide was dried in an oven at 100° C. for 10 hours.

(121) Oxidation Step

(122) The hydroxide obtained in the precipitation step was calcined in the air at 1,000° C. for 1 hour. After cooling, a red SmFeLaW oxide as raw material powder was obtained.

(123) Pretreatment Step

(124) An amount of 100 g of the SmFeLaW oxide obtained in Production Example 2 was put in a steel container to a thickness of 10 mm. The container was placed in a furnace, and the pressure was reduced to 100 Pa. Then, while introducing hydrogen gas, the temperature was increased to a pretreatment temperature of 850° C. and maintained for 15 hours. The oxygen concentration was measured by a non-dispersive infrared (ND-IR) analysis (EMGA-820 available from Horiba Ltd.) and found to be 5% by mass. The results show that a black partial oxide was obtained in which the oxygen bonded to Sm remained unreduced while 95% of the oxygen bonded to Fe was reduced.

(125) Reduction Step

(126) An amount of 60 g of the partial oxide obtained in the pretreatment step was mixed with 19.2 g of metallic calcium having an average particle size of about 6 mm, and the mixture was placed in a furnace. After vacuum evacuation of the furnace, argon gas (Ar gas) was introduced. The temperature was increased to a first temperature of 960° C. and maintained for 45 minutes to obtain SmFeLaW alloy particles.

(127) Nitridation Step

(128) Subsequently, the temperature inside the furnace was lowered to 100° C., followed by vacuum evacuation. Then, while introducing nitrogen gas, the temperature was increased to a first temperature of 430° C. and maintained for 3 hours. Next, the temperature was increased to a second temperature of 500° C. and maintained for 1 hour, followed by cooling to obtain a magnetic particle-containing bulk product.

(129) Post Treatment Step

(130) The bulk product obtained in the nitridation step was introduced into 3 kg of pure water and stirred for 30 minutes. After standing still, the supernatant was drained by decantation. The introduction into pure water, stirring, and decantation were repeated 10 times. Next, 2.5 g of 99.9% acetic acid was introduced and stirred for 15 minutes. After standing still, the supernatant was drained by decantation. The introduction into pure water, stirring, and decantation were repeated twice, followed by solid-liquid separation and then vacuum drying at 80° C. for 3 hours to obtain a SmFeN-based anisotropic magnetic powder.

Example 1

(131) Dispersion Step

(132) The SmFeN-based anisotropic magnetic powder prepared in Production Example 1 and media (nylon-coated iron core media, diameter: 10 mm, Vickers number of nylon coating: 7, specific gravity: 7.48, thickness of nylon layer: about 1 to 3 mm) were put into a container used in a vibration mill so that the amounts of the SmFeN-based anisotropic magnetic powder and the media were 5% by volume and 60% by volume, respectively, relative to the volume of the container. The powder was dispersed by the vibration mill in a nitrogen atmosphere for 60 minutes to obtain a SmFeN-based anisotropic magnetic powder.

Comparative Example 1

(133) The SmFeN-based anisotropic magnetic powder prepared in Production Example 1 and

media (chromium steelballs, SUJ2, diameter: 2.3 mm, Vickers number: 760, specific gravity: 7.77) were put into a container used in a vibration mill so that the amounts of the SmFeN-based anisotropic magnetic powder and the media were 5% by volume and 60% by volume, respectively, relative to the volume of the container. The powder was dispersed by the vibration mill in a nitrogen atmosphere for 60 minutes to obtain a SmFeN-based anisotropic magnetic powder.

Comparative Example 2

(134) The SmFeN-based anisotropic magnetic powder prepared in Production Example 1 and media (nylon material, diameter: 10 mm, Vickers number: 7, specific gravity: 1.13) were put into a container used in a vibration mill so that the amounts of the SmFeN-based anisotropic magnetic powder and the media were 5% by volume and 60% by volume, respectively, relative to the volume of the container. The powder was dispersed by the vibration mill in a nitrogen atmosphere for 60 minutes to obtain a SmFeN-based anisotropic magnetic powder.

Example 2

(135) Dispersion Step

(136) The SmFeN-based anisotropic magnetic powder prepared in Production Example 2 and media (nylon-coated iron core media, diameter: 10 mm, Vickers number of nylon coating: 7, specific gravity: 7.48, thickness of nylon layer: about 1 to 3 mm) were put into a container used in a vibration mill so that the amounts of the SmFeN-based anisotropic magnetic powder and the media were 5% by volume and 60% by volume, respectively, relative to the volume of the container. The powder was dispersed by the vibration mill in a nitrogen atmosphere for 60 minutes to obtain a SmFeN-based anisotropic magnetic powder.

(137) The average particle size, particle size distribution, residual magnetization σ_r , coercive force iH_c , squareness ratio H_k , oxygen concentration, and nitrogen concentration of the SmFeN-based anisotropic magnetic powders obtained in Example 1, Comparative Example 1, Comparative Example 2, and Example 2 were measured as described above, and the results are shown in Table 1. The metal contents of each magnetic powder were measured, and the results are shown in Table 2. Moreover, images of the magnetic powders obtained in Example 1, Comparative Example 1, Comparative Example 2, and Example 2 were taken with a scanning electron microscope (SU3500, Hitachi High-Technologies Corporation, 5 KV, 5,000 $\times$), and the results are shown in FIGS. 1, 2, 3 and 4, respectively.

(138) TABLE-US-00001

TABLE 1	Particle size	O	N	Example	size	D10	D50	D90	Circ	σ_r	iH_c	H_k						
	concentration	concentration	No.	(μm)	(μm)	(μm)	(μm)	Span	(circularity)	(emu/g)	(Oe)	(Oe)	(%)					
(%) Example 1	2.03	0.80	1.79	3.41	1.46	0.836	136.0	13640	4140	0.55	3.30	Comparative	1.90	0.82				
1.76	3.19	1.35	0.833	125.5	13340	5438	0.78	3.40	Example 1	Comparative	2.46	0.94	2.15	4.13	1.48			
0.829	118.6	15440	5045	0.54	3.30	Example 2	Example 2	1.70	0.58	1.39	2.85	1.63	0.831	131.0	13340	5438	0.70	3.32

(139) TABLE-US-00002

TABLE 2	Example	Content (%)	Composition	No.	Sm	Fe	N	W	La	Sm	Fe																																						
	N	W	La	Example 1	24.2	70.8	3.30	1.03	—	9.6	75.9	14.1	0.3	—	Comparative	24.1	71.0	3.40	0.97	—	9.6	76.0	14.5	0.3	—	Example 1	Comparative	24.1	71.2	3.30	1.02	—	9.6	76.0	14.1	0.3	—	Example 2	Example 2	24.3	70.9	3.32	0.95	0.45	9.7	75.8	14.2	0.3	0.2

(140) It was demonstrated that Examples 1 and 2 in which the powders were dispersed using a nylon resin-coated iron core as media had a higher remanence than Comparative Example 1 in which the powder was dispersed using chromium steel balls not coated with a resin as the media and Comparative Example 2 in which the powder was dispersed using a nylon resin as the media. Moreover, the magnetic powder of Comparative Example 1 contained a lot of fine particles as shown in FIG. 2, while the magnetic powders of Examples 1 and 2 contained relatively few fine particles.

(141) The SmFeN-based anisotropic magnetic powder produced by the production method according to embodiments of the present disclosure has a low oxygen concentration and excellent magnetic properties, and thus can be suitably applied to bonded magnets and sintered magnets.

Claims

1. A method of producing a bonded magnet, the method comprising: preparing a SmFeN-based anisotropic magnetic powder before dispersing comprising Sm, Fe, W, and N, and dispersing the SmFeN-based anisotropic magnetic powder using a resin-coated metal media or a resin-coated ceramic media to obtain a SmFeN-based anisotropic magnetic powder, and mixing the SmFeN-based anisotropic magnet powder and a resin, wherein the media has a diameter of at least 2 mm and not more than 100 mm.
2. The method according to claim 1, wherein the media has a specific gravity of not less than 4.
3. The method according to claim 2, wherein the media has a specific gravity of not more than 8.
4. The method according to claim 1, wherein the resin-coated metal media is a nylon resin-coated iron core media.
5. The method according to claim 1, wherein the dispersing is performed by a vibration mill.
6. The method according to claim 5, wherein, in the dispersing, an amount of the media is at least 60% by volume but not more than 70% by volume, and an amount of the SmFeN-based anisotropic magnetic powder before dispersing is at least 3% by volume but not more than 20% by volume, each relative to a volume of a container used to contain the SmFeN-based anisotropic magnetic powder and the media.
7. The method according to claim 1, wherein the dispersing is performed in the absence of a solvent.
8. The method according to claim 1, wherein the dispersing is performed in an inert gas atmosphere.
9. The method according to claim 1, wherein the step of preparing the SmFeN-based anisotropic magnetic powder before dispersing includes: pretreating an oxide comprising Sm, Fe, and W by heat treatment in a reducing gas-containing atmosphere to obtain a partial oxide; heat treating the partial oxide in the presence of a reducing agent to obtain alloy particles; nitriding the alloy particles to obtain a nitride; and washing the nitride to obtain the SmFeN-based anisotropic magnetic powder before dispersing.
10. The method according to claim 1, wherein the SmFeN-based anisotropic magnetic powder has D50 that is a particle size corresponding to 50th percentile of a cumulative particle size distribution by volume of the SmFeN-based anisotropic magnetic powder of at least 0.5 μm but not more than 2.5 μm .
11. A method of producing a bonded magnet, the method comprising: preparing a SmFeN-based anisotropic magnetic powder before dispersing comprising Sm, Fe, W, and N, and dispersing the SmFeN-based anisotropic magnetic powder using a resin-coated metal media or a resin-coated ceramic media to obtain a SmFeN-based anisotropic magnetic powder, and mixing the SmFeN-based anisotropic magnet powder and a resin, wherein the dispersing is performed by a vibration mill, and wherein, in the dispersing, an amount of the media is at least 60% by volume but not more than 70% by volume, and an amount of the SmFeN-based anisotropic magnetic powder before dispersing is at least 3% by volume but not more than 20% by volume, each relative to a volume of a container used to contain the SmFeN-based anisotropic magnetic powder and the media.
12. The method according to claim 11, wherein the media has a specific gravity of not less than 4.
13. The method according to claim 11, wherein the media has a specific gravity of not more than 8.
14. The method according to claim 11, wherein the resin-coated metal media is a nylon resin-coated iron core media.
15. The method according to claim 11, wherein the dispersing is performed in the absence of a solvent.
16. The method according to claim 11, wherein the dispersing is performed in an inert gas

atmosphere.

17. The method according to claim 11, wherein the step of preparing the SmFeN-based anisotropic magnetic powder before dispersing includes: pretreating an oxide comprising Sm, Fe, and W by heat treatment in a reducing gas-containing atmosphere to obtain a partial oxide; heat treating the partial oxide in the presence of a reducing agent to obtain alloy particles; nitriding the alloy particles to obtain a nitride; and washing the nitride to obtain the SmFeN-based anisotropic magnetic powder before dispersing.

18. The method according to claim 11, wherein the SmFeN-based anisotropic magnetic powder has D50 that is a particle size corresponding to 50th percentile of a cumulative particle size distribution by volume of the SmFeN-based anisotropic magnetic powder of at least 0.5 μm but not more than 2.5 μm .
